PRE-ALGEBRA (7TH) · UNIT 10: AREA AND VOLUME

Backyard Pool Makeover

Use area, surface area, and volume to design a backyard with a pool and patio.

CP2

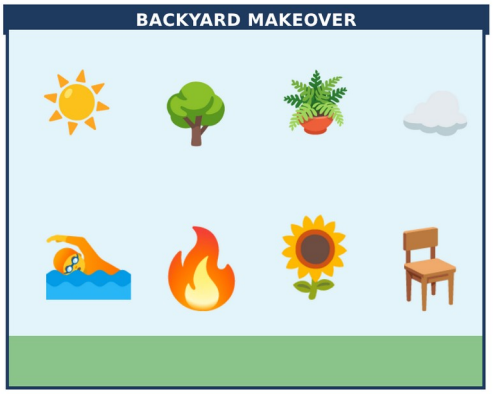
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A family hired your design team to redo their backyard. They want a pool, a stone patio, a garden, and a fire pit, and they have a budget.

You will use area, surface area, and volume to choose shapes, calculate materials, and present your design.



Yard
60 ft by 40 ft

Budget
$25,000

Learning Goals

- I can find the area of parallelograms, triangles, trapezoids, and circles.
- I can find the volume of prisms and cylinders.
- I can use measurements to find costs.

Project Timeline

When	What you will do
Day 1	Patio & Garden
Day 2	The Pool
Day 3	Fire Pit & Cost
Final	Backyard Design Proposal + reflection

Part 1: Patio & Garden

Day 1 · Use the Material Price Cards. Use π ≈ 3.14.

Material Price Cards

Pool (rectangular)	Pool (round)	Patio Stone
30 ft × 15 ft × 5 ft	diameter 24 ft, 4 ft deep	$12 per sq ft
$45 per sq ft of floor	$40 per sq ft of floor	

Pool Liner	Water	Mulch
$2.50 per sq ft	$0.006 per gallon	$40 per cubic yard
	1 cu ft ≈ 7.5 gal	

1. The patio is a parallelogram with base 20 ft and height 12 ft. Find its area.

   Starter: $A = bh$ = ___ × ___

2. Find the cost of patio stone for the patio.

3. A triangle garden has base 10 ft and height 8 ft. Find its area.

4. A trapezoid flower bed has bases 6 ft and 10 ft and height 4 ft. Find its area.

Part 2: The Pool

Day 2 · Compare the two pool shapes.

5. Find the floor area of the rectangular pool (30 ft × 15 ft).

6. Find the floor area of the round pool (diameter 24 ft).

7. Find the volume of water in the rectangular pool (5 ft deep).

8. Find the volume of the round pool (4 ft deep).

Part 3: Fire Pit & Cost

Day 3 · The fire pit is a circle with radius 2 ft.

9. Find the area of the fire pit.

10. How many gallons of water fill the rectangular pool? (1 cu ft ≈ 7.5 gal)

11. Choose a pool and find the cost of its floor. Add the patio. Is it under $25,000?

Final Product: Backyard Design Proposal

Your final product must include:

- A yard drawing with every shape labeled
- Areas of the patio, garden, bed, and fire pit
- Pool volume and gallons
- Cost table
- Your recommendation

Reflection

R1. Why is pool size measured with volume but the patio with area?

R2. How did you use the radius in the circle formulas?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Areas, surface areas, and volumes are computed correctly and used to price and compare designs. Goes beyond the minimum with a creative, realistic choice.	Areas, surface areas, and volumes are computed correctly and used to price and compare designs.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes a labeled drawing, calculations, a cost table, and a recommendation. Polished, organized, and visually engaging.	Proposal includes a labeled drawing, calculations, a cost table, and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.